

Exponential Functions and the Derivative of e^x

MATH 145 – Applied Calculus

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Functions that are the Derivative of Themselves

What functions f have the property that they are the derivative of themselves?

In other words:

$$\frac{d}{dx}f(x) = f(x)$$

The trivial solution is $f(x) = 0$, but that's not very exciting...

Exponential Functions

Exponential Functions

An **exponential function** is a function of the form $f(x) = P_0 a^x$, where P_0 and a are constants.

This is a *family* of functions that depend on P_0 and a

We've seen other families of functions, such as linear functions ($f(x) = mx + b$) or quadratic functions ($f(x) = ax^2 + bx + c$).

Exponential Functions

What is the meaning of P_0 ?

P_0 is the **initial amount**, the value of $f(x)$ when $x = 0$.

What is the meaning of a ? Consider $a = 1 + r$.

r is the **percentage rate of growth** of an exponential function. It is constant over intervals of the same length.

Example: Exponential Functions

Let $f(x) = 1000(1.07)^x$.

1. Find $f(1)$.
2. Find $f(3.5)$.

Example: Exponential Functions

3. Suppose that you start a saving account with \$10,000, that gains 1% interest monthly. How much money will be in the account after 4 years?
4. How much money do I need to put in the account day 1 if I want to ensure I have \$5,000 in the account after 2.5 years?

Graphs of Exponential Functions

The graphs of exponential functions are distinctive because of their monotonic increase/decrease, concavity and asymptotes.

- If $a > 1$, we have **exponential growth**.
- If $0 < a < 1$, we have **exponential decay**.
- If $b > a > 1$, then $b^x > a^x$ for $x > 0$ (and vice versa for $x < 0$).
- If $b < a < 1$, then $b^x < a^x$ for $x > 0$ (and vice versa for $x < 0$).

Laws of Exponents

All variables here represent real numbers and all variables in denominators are nonzero.

$$1. x^a \cdot x^b = x^{a+b}$$

$$2. \frac{x^a}{x^b} = x^{a-b}$$

$$3. (x^a)^b = x^{ab}$$

$$4. (xy)^a = x^a y^a$$

$$5. \left(\frac{x}{y}\right)^b = \frac{x^b}{y^b}$$

$$6. x^0 = 1, \text{ provided } x \neq 0$$

$$7. x^{-a} = \frac{1}{x^a}, \text{ provided } x \neq 0$$

$$8. x^{\frac{1}{n}} = \sqrt[n]{x}, \text{ provided } x \neq 0$$

Example: Laws of Exponents

1. Simplify $(2 \cdot 3^x \cdot 9^x)^{-2}$, and state it in the form of an exponential function.

Continuously Compounding Interest

Consider an account that starts with \$1 and earns an annual interest rate of 100% compounded continuously. How much money will it have in the account?

Continuously Compounding Interest

$$\text{Compounding Yearly: } (1 + 1)^1 = 2$$

$$\text{Compounding Every Six Months: } (1 + \frac{1}{2})^2 = 2.25$$

$$\text{Compounding Every Three Months: } (1 + \frac{1}{4})^4 = 2.4414$$

$$\text{Compounding Monthly: } (1 + \frac{1}{12})^{12} = 2.6130$$

$$\text{Compounding Weekly: } (1 + \frac{1}{52})^{52} = 2.6926$$

$$\text{Compounding Daily: } (1 + \frac{1}{365})^{365} = 2.7146$$

$$\text{Compounding Hourly: } (1 + \frac{1}{8760})^{8760} = 2.7181$$

$$\text{Compounding Every Minute: } (1 + \frac{1}{525600})^{525600} = 2.718279$$

$$\text{Compounding Every Second: } (1 + \frac{1}{31536000})^{31536000} = 2.718282$$

$$\text{Compounding Continuously: } \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e \approx 2.7182818284 \dots$$

e , Euler's number

The most important base for the exponential function is **Euler's number**, $e \approx 2.7182818284\dots$

We call $f(x) = e^x$ the **exponential function**.

In an exponential function of the form $f(x) = P_0 e^{kx}$, k is called the **continuous growth rate**.

$$e^k = \lim_{n \rightarrow \infty} \left(1 + \frac{k}{n}\right)^n$$

Derivative of a^x

What is $\frac{d}{dx} a^x$?

$$\begin{aligned}\frac{d}{dx} a^x &= \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^x a^h - a^x}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^x (a^h - 1)}{h} \\ &= \lim_{h \rightarrow 0} \left(a^x \left(\frac{a^h - 1}{h} \right) \right) \\ &= a^x \left(\lim_{h \rightarrow 0} \left(\frac{a^h - 1}{h} \right) \right)\end{aligned}$$

Derivative of e^x

If $a = e$, then $\lim_{h \rightarrow 0} \left(\frac{a^h - 1}{h} \right) = 1$?

Thus $\frac{d}{dx} e^x = e^x$.

$f(x) = e^x$ is the only function other than the trivial $f(x) = 0$ for which $\frac{d}{dx} f(x) = f(x)$.

Example: Derivative of e^x

1. $\frac{d}{dx} e^{kx}$

2. $\frac{d}{dx} x e^{3x}$

Example: Derivative of e^x

3. $\frac{d}{dx} e^{3x^2+1}$

4. $\frac{d^2}{dx^2} e^{3x^2+1}$

5. $\frac{d}{dx} e^{e^x}$